OPTIMIZATION STUDY

Optimized design candidate

Revision MDO import 2026-10-14 · plate-frame frame · x44-standard motors

Named input case · design_baseline.json

2026-10-15T14:46:39+00:00

All modeled checks pass

63 / 63 physical model checks pass · 6 capabilities not assessed

Passing modeled checks does not establish hardware performance. These are predictions from the stated inputs, not measured performance.

Dry rover mass

42.0 kg

Before adding regolith

Payload per cycle

53.6 kg

Total carried by four drums

Complete cycles

6

Within 15 minutes; no partial delivery credit

Delivered volume

193.1 L

Complete-cycle mission estimate

Construction score

Continuous mission estimate using the configured mass and energy weights. The complete-cycle estimate is reported in Performance & mass.

Construction points

2,345.80

Weighted mass + energy points

Mass points

1,355.17

Dry-mass productivity component

Energy points

990.64

Consumed-energy productivity component

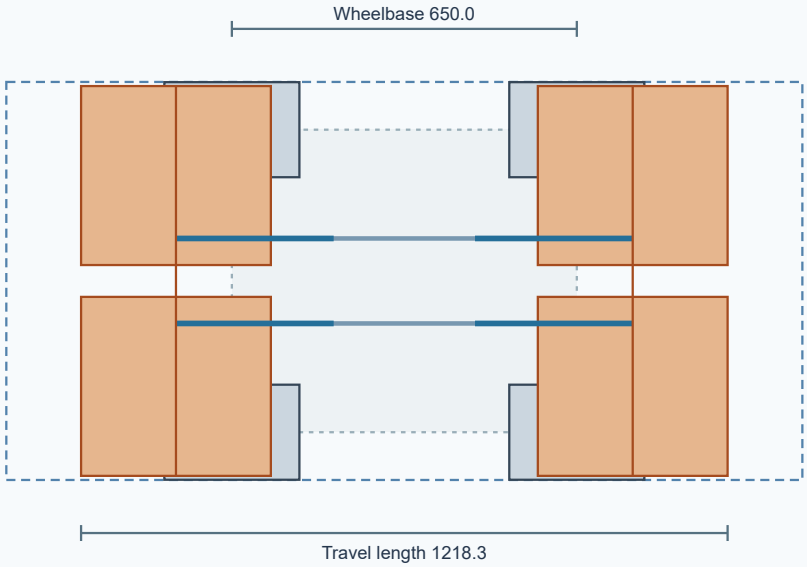
01 / ROVER CONFIGURATION

Geometry at the evaluated inputs

Level-ground model schematics. Check statuses below determine whether envelopes and clearance requirements are satisfied.

Top-down view / travel pose

Orthographic footprints; front is +x (right). Dimensions in mm.



Wheel-center track: 570.1 | Overall width: 749.2 | Wheel width: 179.1

Each drum: 357.6 diameter x 337.3 axial length | Pair gap: 60.0

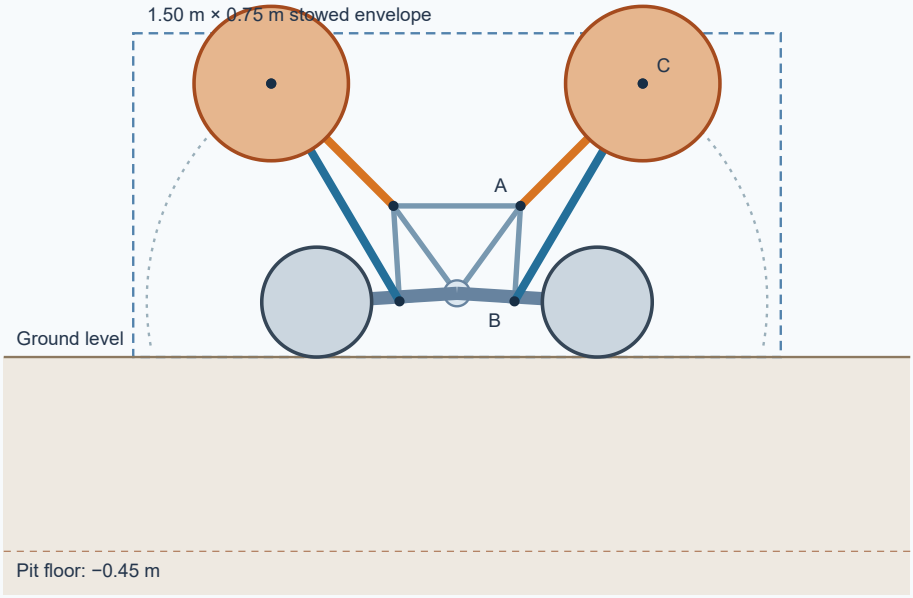
Dashed blue: 1500 x 750 stowed limit (reference only in this travel pose). Gray: wheel-center support polygon.

Projection overlap does not prove collision. Heights, motors, equipment and connection details are not shown.

Travel-pose plan view. Four wheels and four drums are shown at their model stations. Top-down SVG

Raised · travel / deposition

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



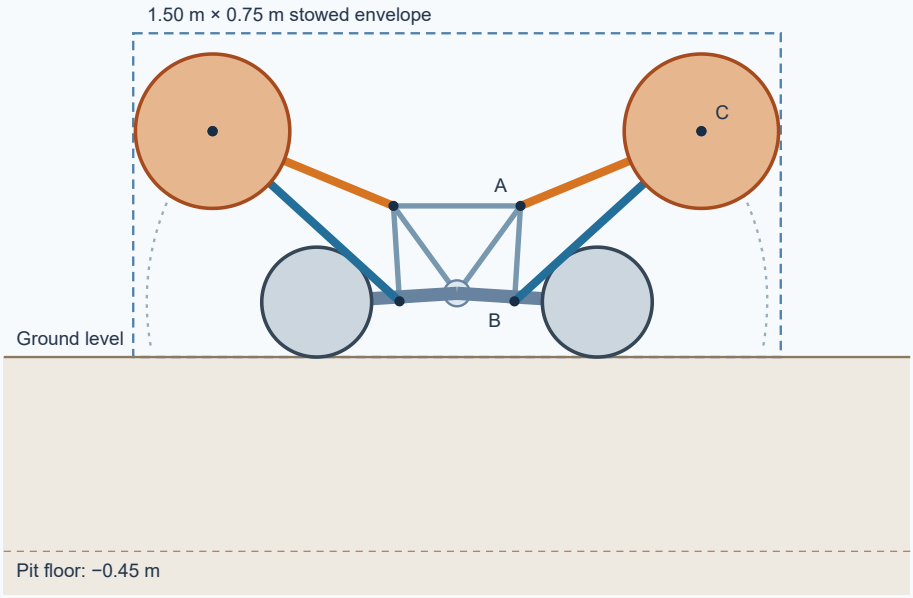
Actuator: 400.6 mm · Fixed arm: 585.5 mm · Drum Ø 357.6 mm

Drum top: +0.812 m · Drum bottom: +0.455 m · A: 0.350 m · B: 0.129 m

Blue = fixed arm · Orange = actuator · Attachment B-D: 585.5 mm (100.0% of arm) · Ground is y = 0

Compact · starting envelope

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



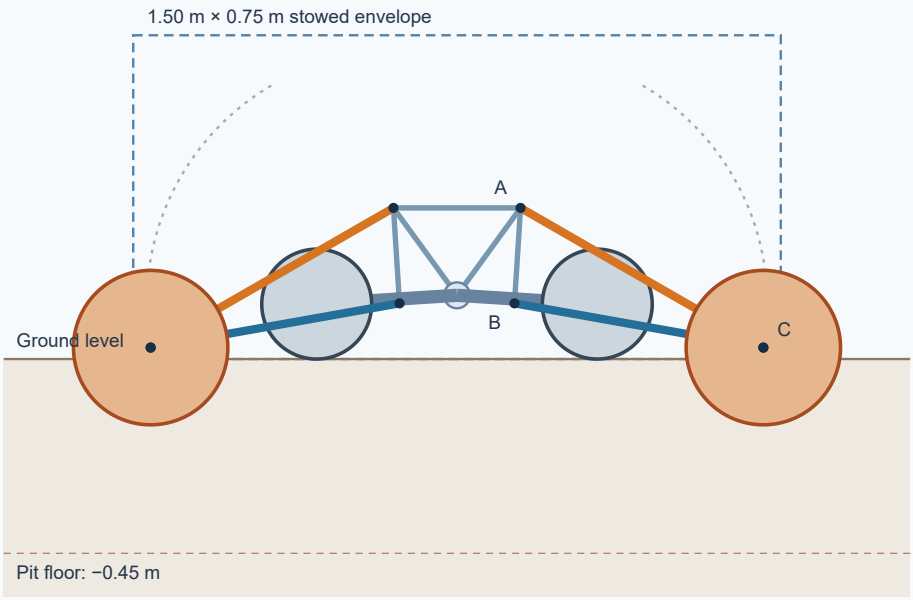
Actuator: 453.3 mm · Fixed arm: 585.5 mm · Drum Ø 357.6 mm

Drum top: +0.702 m · Drum bottom: +0.344 m · A: 0.350 m · B: 0.129 m

Blue = fixed arm · Orange = actuator · Attachment B-D: 585.5 mm (100.0% of arm) · Ground is y = 0

Extended · digging

Side elevation · mirrored front and rear · each visible drum represents a shaft-linked pair



Actuator: 648.9 mm · Fixed arm: 585.5 mm · Drum Ø 357.6 mm

Drum top: +0.205 m · Drum bottom: -0.152 m · A: 0.350 m · B: 0.129 m

Blue = fixed arm · Orange = actuator · Attachment B-D: 585.5 mm (100.0% of arm) · Ground is y = 0

Blue: arm · orange: actuator. Each visible drum represents a shaft-linked pair. Drawings are not CAD.

Plate-frame member schematic

Plate frame: provisional member topology

Lengths and directions follow the mounting coordinates; this is not the manufactured cutout outline.

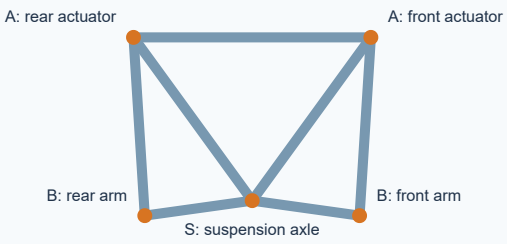


Plate thickness: 6.35 mm (fixed)

Common strip width: 12.49 mm

CF ID / OD: 25.40 / 27.40 mm

Plate spacing / CF span: 160.2 mm

Rocker bearing spacing: 345.3 mm

Wheel-to-rocker gap: 12.7 mm

Central support: pin, balanced mirrored gravity only.

CF material properties are PROVISIONAL.

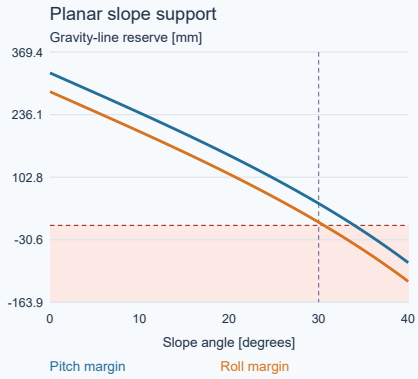
Spoke buckling is a strip screen,

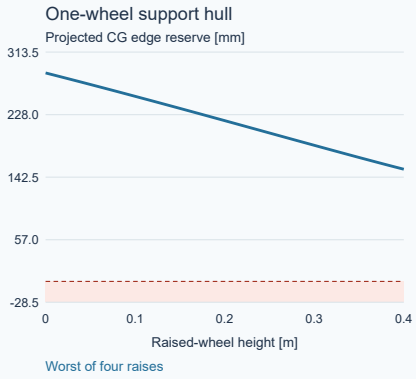
not whole-plate buckling validation.

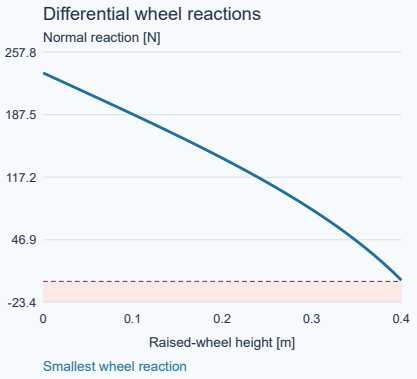
Static slope and articulation curves

Loaded travel pose: gravity stability screens

One wheel raised at a time on horizontal platforms. Curves are model calculations; no physical rock profile is simulated.







02 / PREDICTED OUTPUT

Performance & mass

Complete-cycle estimates and continuous MDO estimates are labeled separately.

Prediction	Value	Meaning
Full cycle duration	149.4 s	Includes digging, actuation, travel, dumping and relocation; overlap removed once.
Energy per cycle	3.25 Wh	Electrical energy including auxiliaries.
Mission energy · complete cycles	19.5 Wh	Includes idle electronics for remaining mission time.
Mission energy · continuous estimate	19.6 Wh	Used by the existing battery constraint and MDO objective.
Peak electrical power	671.3 W	Largest modeled simultaneous demand; not an average power rating.
Loaded mass	95.6 kg	Dry mass + all four drums' regolith payload.
Actual route peak speed	0.57 m/s	Prescribed motion, not measured vehicle performance.
Score · complete-cycle estimate	2,339.10 points	Uses complete deliveries and idle remainder; separate from the continuous MDO objective.

Dry mass breakdown

Each assembly is counted once.

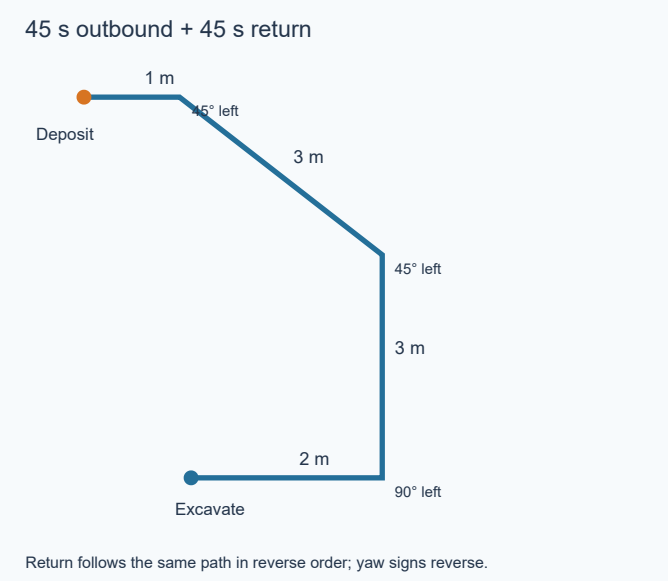
Equipment allowance	6.00 kg
Frame	4.28 kg
Four bare wheels	7.84 kg
Four drive assemblies	4.15 kg
Four drums + two drive assemblies	13.8 kg
Battery	0.34 kg
Arms + actuators	5.58 kg

Electrical energy per cycle

Work / provisional efficiency, plus auxiliaries.

Digging	1.02 Wh
Dumping	0.08 Wh
Travel + deposition turn	0.83 Wh
Actuators	0.22 Wh
Fresh-site move	0.07 Wh
Electronics	1.04 Wh

Route & cycle accounting



45 s outbound + 45 s return

Dig and lower concurrently → raise loaded drums → outbound → dump one pair → turn → dump second pair → return → relocate to fresh ground.

Lowering overlap is subtracted once. The deposition turn and fresh-site relocation are additional to the timed route. Modeled full cycle: 149.4 s.

03 / GAPS TO RESOLVE

Requirements & model checks

Failures are listed first. “Not assessed” means the model does not verify that capability.

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Dry rover mass `dry_mass`	42.0 kg	≤ 80.0 kg	38.0 kg	Pass
Compact overall length `rover_length`	1,490 mm	≤ 1,500 mm	10.1 mm	Pass
Overall width `rover_width`	749.2 mm	≤ 750.0 mm	0.81 mm	Pass
Compact overall height `rover_height`	701.9 mm	≤ 750.0 mm	48.1 mm	Pass
Excavation shaft-power reserve `drum_power_margin`	Reserve: 1,002 W	Reserve ≥ 0	1,002 W	Pass
Excavation: penetration reserve `penetration_margin`	Reserve: 391.4 N	Reserve ≥ 0	391.4 N	Pass
Excavation: traction reserve `dig_traction_margin`	Reserve: 228.8 N	Reserve ≥ 0	228.8 N	Pass
Cut depth margin `cut_depth_margin`	Reserve: 6.41 mm	Reserve ≥ 0	6.41 mm	Pass
Ground contact pressure `ground_pressure_margin`	13.2 kPa	≤ 25.0 kPa	11.8 kPa	Pass
Straight travel: motor-force reserve `straight_motor_margin`	Reserve: 4,487 N	Reserve ≥ 0	4,487 N	Pass
Turning: motor-force reserve `turn_motor_margin`	Reserve: 6,657 N	Reserve ≥ 0	6,657 N	Pass
Straight travel: traction reserve `straight_traction_margin`	Reserve: 210.3 N	Reserve ≥ 0	210.3 N	Pass
Turning: traction reserve `turn_traction_margin`	Reserve: 10.2 N	Reserve ≥ 0	10.2 N	Pass
Straight travel: shaft-power reserve `straight_power_margin`	Reserve: 2,612 W	Reserve ≥ 0	2,612 W	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Turning: shaft-power reserve `turn_power_margin`	Reserve: 2,609 W	Reserve ≥ 0	2,609 W	Pass
Battery usable energy `battery_energy_margin`	19.6 Wh	≤ 37.9 Wh	18.3 Wh	Pass
Battery peak-power capacity `battery_power_margin`	671.3 W	≤ 710.2 W	38.9 W	Pass
Time remaining after one complete cycle `complete_cycle_margin`	Reserve: 750.6 s	Reserve ≥ 0	750.6 s	Pass
Minimum digging-depth reserve `dig_reach_margin`	Reserve: 147.3 mm	Reserve ≥ 0	147.3 mm	Pass
Pit depth margin `pit_depth_margin`	Reserve: 297.7 mm	Reserve ≥ 0	297.7 mm	Pass
Transport clearance margin `transport_clearance_margin`	Reserve: 264.3 mm	Reserve ≥ 0	264.3 mm	Pass
Travel drum height above wheel-diameter step `wheel_first_margin`	Reserve: 200.2 mm	Reserve ≥ 0	200.2 mm	Pass
Startup position margin `startup_position_margin`	Reserve: 52.7 mm	Reserve ≥ 0	52.7 mm	Pass
Dig position margin `dig_position_margin`	Reserve: 195.6 mm	Reserve ≥ 0	195.6 mm	Pass
Pivot spacing margin `pivot_spacing_margin`	Reserve: 141.5 mm	Reserve ≥ 0	141.5 mm	Pass
Pivot mount margin `pivot_mount_margin`	Reserve: 254.5 mm	Reserve ≥ 0	254.5 mm	Pass
Pivot longitudinal margin `pivot_longitudinal_margin`	Reserve: 191.8 mm	Reserve ≥ 0	191.8 mm	Pass
Actuator pivot longitudinal margin `actuator_pivot_longitudinal_margin`	Reserve: 177.8 mm	Reserve ≥ 0	177.8 mm	Pass
Bottom mount margin `bottom_mount_margin`	Reserve: 1.75 mm	Reserve ≥ 0	1.75 mm	Pass
Stroke margin `stroke_margin`	Reserve: 208.3 mm	Reserve ≥ 0	208.3 mm	Pass
Actuator body margin `actuator_body_margin`	Reserve: 112.4 mm	Reserve ≥ 0	112.4 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Triangle retracted margin `triangle_retracted_margin`	Reserve: 21.6 mm	Reserve ≥ 0	21.6 mm	Pass
Triangle extended margin `triangle_extended_margin`	Reserve: 143.1 mm	Reserve ≥ 0	143.1 mm	Pass
Toggle margin `toggle_margin`	Reserve: 282.2 mm	Reserve ≥ 0	282.2 mm	Pass
Actuator toggle margin `actuator_toggle_margin`	Reserve: 249.6 mm	Reserve ≥ 0	249.6 mm	Pass
Wheel clearance margin `wheel_clearance_margin`	Reserve: 68.2 mm	Reserve ≥ 0	68.2 mm	Pass
Deployed height margin `deployed_height_margin`	Reserve: 1,688 mm	Reserve ≥ 0	1,688 mm	Pass
Actuator force capacity `actuator_force_margin`	1,252 N	≤ 1,335 N	83.2 N	Pass
Excavation torque reserve `drum_torque_margin`	Reserve: 74.0 N·m	Reserve ≥ 0	74.0 N·m	Pass
Dumping torque reserve `dump_torque_margin`	Reserve: 81.9 N·m	Reserve ≥ 0	81.9 N·m	Pass
Minimum payload reserve `payload_margin`	Reserve: 53.6 kg	Reserve ≥ 0	53.6 kg	Pass
Excavated trench diameter reserve `trench_traverse_margin`	Reserve: 102.2 mm	Reserve ≥ 0	102.2 mm	Pass
Fresh site margin `fresh_site_margin`	Reserve: 65.5 mm	Reserve ≥ 0	65.5 mm	Pass
Frame section margin `frame_section_margin`	Reserve: 4.02 mm	Reserve ≥ 0	4.02 mm	Pass
Arm section margin `arm_section_margin`	Reserve: 5.53 mm	Reserve ≥ 0	5.53 mm	Pass
Structural stress envelope `structure_stress_margin`	114.6 MPa	≤ 120.0 MPa	5.37 MPa	Pass
Structural deflection envelope `structure_deflection_margin`	3.80 mm	≤ 5.00 mm	1.20 mm	Pass
Structural buckling reserve `structure_buckling_margin`	Reserve: 168.8 N	Reserve ≥ 0	168.8 N	Pass
Level-route dynamic tip reserve `mobility_stability_margin`	Reserve: 272.3 mm	Reserve ≥ 0	272.3 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Asymmetric dump tip reserve `tip_stability_margin`	Reserve: 48.3 mm	Reserve ≥ 0	48.3 mm	Pass
Rock pose: geometric reach reserve `obstacle_pose_margin`	Reserve: 173.3 mm	Reserve ≥ 0	173.3 mm	Pass
Rock endpoint: global tip reserve `obstacle_tip_margin`	Reserve: 153.4 mm	Reserve ≥ 0	153.4 mm	Pass
Obstacle drum–wheel clearance `obstacle_drum_wheel_margin`	16.6 mm	≥ 15.0 mm	1.58 mm	Pass
30° slope: static tip reserve `slope_tip_margin`	Reserve: 6.70 mm	Reserve ≥ 0	6.70 mm	Pass
Rock path: global tip reserve `rock_sweep_tip_margin`	Reserve: 153.4 mm	Reserve ≥ 0	153.4 mm	Pass
Rock path: smallest wheel reaction `rock_sweep_wheel_load_margin`	Reserve: 1.37 N	Reserve ≥ 0	1.37 N	Pass
Available excavation area `excavation_area_margin`	Reserve: 4.14 m^2	Reserve ≥ 0	4.14 m^2	Pass
Fresh-site move: drive/traction reserve `site_drive_margin`	Reserve: 155.7 N	Reserve ≥ 0	155.7 N	Pass
Plate geometry reserve `plate_geometry_margin`	Reserve: 37.7 mm	Reserve ≥ 0	37.7 mm	Pass
Frame packaging reserve `frame_packing_margin`	Reserve: 79.4 mm	Reserve ≥ 0	79.4 mm	Pass
Plate strip stress `plate_stress_margin`	11.1 MPa	≤ 120.0 MPa	108.9 MPa	Pass
Plate strip buckling reserve `plate_buckling_margin`	Reserve: 1,028 N	Reserve ≥ 0	1,028 N	Pass
CF tube bending stress `cf_bending_margin`	109.3 MPa	≤ 175.0 MPa	65.7 MPa	Pass
Wheel diameter search range `wheel_diameter_bounds`	254.5 mm	180.0 mm to 359.0 mm	74.5 mm	Pass
Wheel width search range `wheel_width_bounds`	179.1 mm	50.0 mm to 359.0 mm	129.1 mm	Pass
Drum outer diameter search range `drum_diameter_bounds`	357.6 mm	160.0 mm to 359.0 mm	1.39 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Width of each drum search range `drum_length_bounds`	337.3 mm	120.0 mm to 359.0 mm	21.7 mm	Pass
Excavation speed search range `drum_rpm_bounds`	120.0 rpm	5.00 rpm to 120.0 rpm	0.04 rpm	Pass
Dumping speed search range `dump_rpm_bounds`	60.0 rpm	5.00 rpm to 60.0 rpm	0.00351 rpm	Pass
Drum reduction ratio search range `drum_gear_ratio_bounds`	39.9	9.00 to 125.0	30.9	Pass
Scoop protrusion / cut depth search range `cut_depth_bounds`	11.1 mm	3.00 mm to 25.0 mm	8.09 mm	Pass
Arm tube diameter search range `arm_tube_diameter_bounds`	26.2 mm	25.0 mm to 100.0 mm	1.17 mm	Pass
Arm wall search range `arm_wall_bounds`	1.01 mm	1.00 mm to 5.00 mm	0.01 mm	Pass
Battery capacity search range `battery_capacity_bounds`	47.3 Wh	10.0 Wh to 200.0 Wh	37.3 Wh	Pass
Actuator speed search range `actuator_speed_bounds`	0.07 m/s	0.005 m/s to 0.08 m/s	0.01 m/s	Pass
Travel actuator length search range `travel_actuator_length_bounds`	400.6 mm	300.0 mm to 650.0 mm	100.6 mm	Pass
Compact actuator length search range `actuator_retracted_length_bounds`	453.3 mm	250.0 mm to 650.0 mm	196.7 mm	Pass
Digging actuator length search range `actuator_extended_length_bounds`	648.9 mm	450.0 mm to 1,050 mm	198.9 mm	Pass
Actuator pivot height search range `top_pivot_height_bounds`	350.1 mm	350.0 mm to 700.0 mm	0.06 mm	Pass
Actuator pivot longitudinal position search range `actuator_pivot_x_bounds`	147.2 mm	50.0 mm to 300.0 mm	97.2 mm	Pass
Arm pivot height search range `bottom_pivot_height_bounds`	129.0 mm	90.0 mm to 300.0 mm	39.0 mm	Pass
Arm length search range `arm_length_bounds`	585.5 mm	300.0 mm to 700.0 mm	114.5 mm	Pass

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Arm pivot longitudinal position search range `pivot_x_bounds`	133.2 mm	50.0 mm to 300.0 mm	83.2 mm	Pass
Wheelbase search range `wheelbase_bounds`	650.0 mm	380.0 mm to 650.0 mm	5.97e-05 mm	Pass
Wheel-center track search range `track_width_bounds`	570.1 mm	500.0 mm to 699.0 mm	70.1 mm	Pass
Actuator rating, each search range `actuator_force_rating_bounds`	2,002 N	500.0 N to 6,000 N	1,502 N	Pass
CF tube span / plate spacing search range `cf_tube_span_bounds`	160.2 mm	160.0 mm to 460.0 mm	0.16 mm	Pass
CF wall thickness search range `cf_tube_wall_bounds`	1.00 mm	1.00 mm to 6.00 mm	2.56e-05 mm	Pass
Plate strip width search range `plate_spoke_width_bounds`	12.5 mm	12.0 mm to 70.0 mm	0.49 mm	Pass
Rocker box width search range `rocker_box_width_bounds`	20.3 mm	20.0 mm to 65.0 mm	0.25 mm	Pass
Rocker box height search range `rocker_box_height_bounds`	30.7 mm	25.0 mm to 90.0 mm	5.74 mm	Pass
Rocker wall search range `rocker_wall_bounds`	1.05 mm	1.00 mm to 5.00 mm	0.05 mm	Pass
Axle outer diameter search range `axle_outer_diameter_bounds`	33.1 mm	25.0 mm to 90.0 mm	8.06 mm	Pass
Axle wall search range `axle_wall_bounds`	1.01 mm	1.00 mm to 6.00 mm	0.01 mm	Pass
Rocker pivot above wheel centers search range `rocker_pivot_offset_bounds`	20.3 mm	20.0 mm to 150.0 mm	0.35 mm	Pass
30° slope climbing / sliding	The slope check assumes enough friction and holding torque. Uphill drive capability and sideways sliding are not evaluated.	—	—	Not assessed

Check	Prediction / reserve	Limit / criterion	Signed margin	Status
Real 40 cm rock traversal	Static poses and clearance screens do not establish the driving forces or actual terrain-contact sequence.	—	—	Not assessed
50 cm ditch crossing	Ditch width/radius and contact sequence have not been defined.	—	—	Not assessed
Continuous motor operation	Stator-current limits and thermal duty remain undecided.	—	—	Not assessed
Connections & material validation	Bonded inserts, bushings, bolts, local failure and selected CF tube properties need separate checks.	—	—	Not assessed
Soil and skid-steer predictions	Rolling, traction, capture and scrub relationships remain uncalibrated.	—	—	Not assessed

Positive margins satisfy the modeled criterion. Status uses the saved model's numerical tolerance. Search ranges are hidden by default and available in the filter; they remain part of saved feasibility. Margins use their own units and cannot be ranked directly across unlike checks.

Loads & structural response

Values identify their load case and whether they apply per component or to the whole rover.

Frame gravity solution is valid for its declared load case. Structural load factor: 1.5; structural safety factor: 2.

Quantity / scope	Prediction	Load case / interpretation
Required force per actuator `required_actuator_force`	1,252 N	Loaded raising/extraction envelope, including the model's breakout allowance.
Frame gravity reaction at each actuator `frame_gravity_actuator_force`	1,442 N	Mirrored loaded gravity sweep; includes the structural load factor. Separate from actuator sizing load.
Nominal force per wheel `frame_nominal_wheel_reaction`	234.4 N	Equal-share level-ground weight. Not a rock or slope wheel-load envelope.
Total cutting force `cut_force`	40.4 N	Across four drums at the selected bite and engagement.
Worst modeled contact pressure `ground_pressure`	13.2 kPa	Route pressure estimate, including load transfer and rocker factor.
Structural stress envelope `structure_stress`	114.6 MPa	Envelope of local member stresses; not one assembled-frame FEA result.
Structural deflection envelope `structure_deflection`	3.80 mm	Conservative combination of local/relative deflections.
Upper/lower mount relative displacement `frame_mount_relative_deflection`	0.49 mm	Frame gravity sweep.
Rocker-end flexure `rocker_deflection`	3.80 mm	Local beam-screen displacement.
Suspension-axle flexure `axle_deflection`	0.18 mm	Axle-overhang screen.
Arm-tip flexure `arm_deflection`	0.04 mm	Relative arm beam displacement.
Loaded CG height `loaded_cg_z`	482.1 mm	Fully loaded travel pose, above level ground.

Model notes & engineering data

Keep assumptions and validity limits attached to the predictions.

Obstacle and structural interpretation

Requested endpoint rock pose is geometrically reachable; this alone does not establish traversal.

Obstacle drum–wheel clearance

Requested-height obstacle clearance screen. Spheres enclose the drum and wheel envelopes. A negative gap bound is conservatively rejected; it does not prove actual cylinder collision.

Ditch requirement remains unverified. Static slope tipping assumes adequate friction; it does not verify climbing. The 5 mm structural limit applies to the modeled envelope of local deflections, not one assembled-frame displacement.

For the plate frame, CF tube stiffness/strength are PROVISIONAL. Connections, dynamic obstacle loads, cutting/recovery frame cases and whole-plate buckling need separate verification.

Standard X44 motors · assumptions

Four wheel motors at 100:1; two drum motors at 39.87:1 (exploratory gearing).

Electrical energy = output-shaft mechanical work / combined efficiency. Drive: 72%; drums: 70%. Both efficiencies are provisional.

Continuous operation unverified. Current-limit inputs of zero mean undecided. Shaft torque-speed capability is enforced separately from energy accounting.

Removed screens: diagnostics only

- `dig_froude_margin`: User removed the uncalibrated speed cap; constant capture/lift response assumes no additional high-speed loss.
- `dump_froude_margin`: User removed the uncalibrated speed cap; actual dumping/retention speed response remains unverified.
- `chassis_clearance_margin`: Legacy filled rectangular envelope does not represent the open plate frame; actual frame/equipment interference remains unverified.

06 / MDO ONLY

Optimization study

The candidate is separate from a released design baseline. Optimizer status: **True**. Saved status: converged_feasible.

Continuous-score objective: 2,265.04 → 2,345.80 points. Complete-cycle estimate: 2,339.10 points.

Changed input	Starting value	Candidate value
Wheel diameter	255.0 mm	254.5 mm
Wheel width	176.0 mm	179.1 mm
Drum outer diameter	359.0 mm	357.6 mm
Width of each drum	340.0 mm	337.3 mm
Excavation speed	120.0 rpm	120.0 rpm
Dumping speed	60.0 rpm	60.0 rpm
Drum reduction ratio	40.0	39.9
Scoop protrusion / cut depth	11.0 mm	11.1 mm
Arm tube diameter	26.0 mm	26.2 mm
Arm wall	1.00 mm	1.01 mm
Battery capacity	46.5 Wh	47.3 Wh
Actuator speed	0.07 m/s	0.07 m/s
Travel actuator length	405.0 mm	400.6 mm
Compact actuator length	454.0 mm	453.3 mm
Digging actuator length	651.0 mm	648.9 mm
Actuator pivot height	350.0 mm	350.1 mm
Actuator pivot longitudinal position	150.0 mm	147.2 mm
Arm pivot height	130.0 mm	129.0 mm
Arm length	590.0 mm	585.5 mm
Arm pivot longitudinal position	137.0 mm	133.2 mm
Wheelbase	650.0 mm	650.0 mm
Wheel-center track	570.0 mm	570.1 mm
Actuator rating, each	2,000 N	2,002 N
CF tube span / plate spacing	160.0 mm	160.2 mm
CF wall thickness	1.00 mm	1.00 mm
Plate strip width	12.5 mm	12.5 mm
Rocker box width	20.0 mm	20.3 mm
Rocker box height	30.0 mm	30.7 mm
Rocker wall	1.00 mm	1.05 mm
Axle outer diameter	33.0 mm	33.1 mm
Axle wall	1.00 mm	1.01 mm
Rocker pivot above wheel centers	20.0 mm	20.3 mm

APPENDIX / NATIVE UNITS

Design inputs & output data

140 inputs supplied by file; 0 CLI overrides; 0 remaining inputs use model defaults. Optimizer-selected design values appear in the bounds table. Values below retain their native model units.

All effective inputs & model assumptions

Includes mission, soil, material properties and inactive legacy inputs. Source SHA-256: `d240bca2c1474879c65dc87e0284230eeb933d88bef51697b111eedfb76b036b`.

Input port	Effective value	Native units
wheel_diameter	0.25451879	m
wheel_width	0.17912918	m
drum_diameter	0.35761266	m
drum_length	0.33725535	m
drum_rpm	119.96178	rpm
dump_rpm	59.996489	rpm
drum_gear_ratio	39.867903	—
cut_depth	0.011085549	m
arm_tube_diameter	0.026170902	m
arm_wall	0.0010124527	m
battery_capacity	47.345336	W*h
actuator_speed	0.06635036	m/s
travel_actuator_length	0.40062649	m
actuator_retracted_length	0.45330322	m
actuator_extended_length	0.64887869	m
top_pivot_height	0.3500647	m
actuator_pivot_x	0.14718697	m
bottom_pivot_height	0.12900497	m
arm_length	0.58550046	m
pivot_x	0.13317335	m
wheelbase	0.64999994	m
track_width	0.57006559	m
actuator_force_rating	2002.2703	N
cf_tube_span	0.16015909	m
cf_tube_wall	0.0010000256	m
plate_spoke_width	0.012488203	m
rocker_box_width	0.020251267	m
rocker_box_height	0.030742973	m
rocker_wall	0.0010451694	m

Input port	Effective value	Native units
axle_outer_diameter	0.033061589	m
axle_wall	0.0010101215	m
rocker_pivot_offset	0.020348501	m
actuator_attachment_fraction	1	—
gravity	9.80665	m/s**2
run_time	900	s
traversal_time_one_way	45	s
travel_speed	1	m/s
acceleration	0.4	m/s**2
turn_angle	3.1415927	rad
overhead_time	8	s
electronics_power	25	W
actuator_idle_power	2	W
soil_density	1500	kg/m**3
berm_density	1500	kg/m**3
fill_fraction	0.4	—
delivery_efficiency	0.9	—
capture_fraction	0.65	—
cut_pressure	18000	Pa
cut_engagement	0.15	—
internal_lift_factor	0.15	—
vertical_cut_fraction	0.5	—
force_imbalance	0.15	—
dump_revolutions	3	—
drum_efficiency	0.7	—
drum_gear_efficiency	0.8	—
drive_efficiency	0.72	—
drive_gear_efficiency	0.8	—
motor_rated_rpm	3000	rpm
traction_coefficient	0.6	—
rocker_load_factor	1.2	—
wheel_deflection_fraction	0.07	—
rolling_base	0.04	—
rolling_pressure_scale	200000	Pa
allowable_ground_pressure	25000	Pa
scrub_coefficient	0.25	—
turn_speed_fraction	0.35	—
turn_accel_fraction	0.5	—
chassis_height	0.35	m
drum_pair_gap	0.06	m
fixed_mass	6	kg

Input port	Effective value	Native units
wheel_areal_mass	8	kg/m**2
drum_areal_mass	5	kg/m**2
motor_mass_per_watt	0.006	kg/W
gear_mass_per_torque	0.05	kg/(N*m)
drive_fixed_mass	0.35	kg
drum_fixed_mass	0.5	kg
battery_specific_energy	140	W*h/kg
battery_usable_fraction	0.8	—
battery_c_rate	15	1/h
mass_score_weight	1	—
energy_score_weight	1	—
transport_clearance	0.08	m
mechanism_clearance	0.015	m
minimum_pivot_spacing	0.08	m
minimum_stroke	0.04	m
actuator_housing_allowance	0.04	m
pit_depth	0.45	m
actuator_efficiency	0.65	—
actuator_safety_factor	1.5	—
breakout_force	100	N
actuator_fixed_mass	0.6	kg
actuator_mass_per_force	0.0004	kg/N
actuator_mass_per_power	0.003	kg/W
actuator_mass_per_length	1	kg/m
stored_soil_density	1250	kg/m**3
usable_volume_fraction	0.9	—
scoop_opening	0.035	m
scoop_depth_fraction	0.5	—
scoops_per_revolution	4	—
max_dig_froude	0.3	—
max_dump_froude	0.2	—
drum_spinup_time	2	s
drum_torque_service_factor	1.25	—
minimum_dig_depth	0.005	m
minimum_payload	0.05	kg
fresh_site_distance	0.8	m
site_speed	0.1	m/s
available_excavation_area	12	m**2
tube_density	2700	kg/m**3
tube_youngs_modulus	6.9e+10	Pa
tube_yield_strength	2.4e+08	Pa

Input port	Effective value	Native units
structure_safety_factor	2	—
structure_load_factor	1.5	—
allowable_deflection	0.005	m
joint_mass	2	kg
differential_mass	1	kg
obstacle_height	0.4	m
slope_angle	0.52359878	rad
frame_tube_diameter	0.055	m
frame_wall	0.002	m
plate_thickness	0.00635	m
cf_tube_inner_diameter	0.0254	m
wheel_rocker_gap	0.0127	m
plate_hub_radius	0.03	m
plate_cutout_clearance	0.005	m
frame_mount_clearance	0.003	m
minimum_axle_overhang	0.01	m
plate_buckling_k	1	—
cf_density	1600	kg/m**3
cf_flexural_modulus	7e+10	Pa
cf_flexural_strength	3.5e+08	Pa
drive_motor_power	835.4	W
drum_motor_power	835.4	W
drive_gear_ratio	100	—
motor_bus_voltage	12	V
x44_motor_mass	0.34019428	kg
drive_gearbox_mass	0.697	kg
drum_gearbox_mass	0.753	kg
drive_stator_current_limit	0	A
drum_stator_current_limit	0	A

Design variables and bounds

Optimizer search ranges and evaluated design variables. These ranges guide MDO and input validation; they are not all hardware requirements.

Port	Lower	Evaluated	Upper	Native units
wheel_diameter	0.18	0.254519	0.359	m
wheel_width	0.05	0.179129	0.359	m
drum_diameter	0.16	0.357613	0.359	m
drum_length	0.12	0.337255	0.359	m
drum_rpm	5	119.962	120	rpm
dump_rpm	5	59.9965	60	rpm
drum_gear_ratio	9	39.8679	125	—
cut_depth	0.003	0.0110855	0.025	m
arm_tube_diameter	0.025	0.0261709	0.1	m
arm_wall	0.001	0.00101245	0.005	m
battery_capacity	10	47.3453	200	W*h
actuator_speed	0.005	0.0663504	0.08	m/s
travel_actuator_length	0.3	0.400626	0.65	m
actuator_retracted_length	0.25	0.453303	0.65	m
actuator_extended_length	0.45	0.648879	1.05	m
top_pivot_height	0.35	0.350065	0.7	m
actuator_pivot_x	0.05	0.147187	0.3	m
bottom_pivot_height	0.09	0.129005	0.3	m
arm_length	0.3	0.5855	0.7	m
pivot_x	0.05	0.133173	0.3	m
wheelbase	0.38	0.65	0.65	m
track_width	0.5	0.570066	0.699	m
actuator_force_rating	500	2002.27	6000	N
cf_tube_span	0.16	0.160159	0.46	m
cf_tube_wall	0.001	0.00100003	0.006	m
plate_spoke_width	0.012	0.0124882	0.07	m
rocker_box_width	0.02	0.0202513	0.065	m
rocker_box_height	0.025	0.030743	0.09	m
rocker_wall	0.001	0.00104517	0.005	m
axle_outer_diameter	0.025	0.0330616	0.09	m
axle_wall	0.001	0.00101012	0.006	m
rocker_pivot_offset	0.02	0.0203485	0.15	m

All output ports

Output port	Value	Native units
dry_mass	41.969048	kg
payload_mass	53.642895	kg
wheel_mass	7.8395705	kg
drum_mass	13.782019	kg
drive_mass	4.1487771	kg
battery_mass	0.33818097	kg
rover_length	1.4899079	m
rover_width	0.74919478	m
rover_height	0.70194841	m
ground_pressure	13189.798	Pa
actual_travel_speed	0.57463892	m/s
dig_time	15.143526	s
travel_time	90	s
dump_time	10.000351	s
turn_time	4.3084484	s
cycle_time	149.39843	s
cycle_energy	3.2489849	W*h
cycles	6.0241597	—
berm_volume	0.19389202	m**3
run_energy	19.572404	W*h
average_power	78.289616	W
peak_power	671.31236	W
mass_points	1355.1652	—
energy_points	990.63978	—
construction_points	2345.805	—
no_load_speed	1.0338228	m/s
loaded_motor_speed_limit	1.0231463	m/s
loaded_power_speed_limit	25.026678	m/s
stall_drive_force	10343.239	N
actuation_time	7.4830703	s
arm_energy	779.35502	J
arm_peak_power	259.53504	W
required_actuator_force	1251.672	N
linkage_mass	5.578899	kg
stroke	0.2482522	m
stowed_x	0.56614762	m
stowed_y	0.52314208	m

Output port	Value	Native units
travel_x	0.43032871	m
travel_y	0.63349429	m
extended_x	0.70962867	m
extended_y	0.026486442	m
sweep_x_min	0.43032871	m
sweep_x_max	0.7186738	m
actuator_leverage_max	2.9025857	—
actuator_attachment_distance	0.58550046	m
chassis_clearance_margin	-0.089009023	m
stowed_length	1.4899079	m
stowed_height	0.70194841	m
deployed_length	1.794938	m
deployed_height	0.81230062	m
route_distance	18	m
route_turn_angle	360	deg
route_translation_time	64.856079	s
route_turn_time	25.143921	s
route_peak_acceleration	0.13208395	m/s**2
payload_volume	0.042914316	m**3
fresh_volume	0.055018354	m**3
storage_capacity	0.042918541	m**3
reachable_soil_mass	53.642895	kg
target_payload_mass	53.648176	kg
trench_depth	0.15231989	m
excavation_footprint	1.30513	m**2
cut_force	40.377534	N
drum_froude	2.8774331	—
dig_overlap_time	3.7415351	s
dig_power	386.77732	W
dump_power	65.318912	W
dig_energy	3670.5801	J
dump_energy	270.74944	J
drum_power_margin	1001.7044	W
drum_torque_margin	73.961372	N*m
dump_torque_margin	81.923197	N*m
dig_froude_margin	-2.5774331	—
dump_froude_margin	-0.5197325	—
penetration_margin	391.387	N
dig_traction_margin	228.77557	N
cut_depth_margin	0.0064144514	m
dig_reach_margin	0.14731989	m

Output port	Value	Native units
payload_margin	53.592895	kg
structure_mass	4.2816016	kg
rocker_mass	0.35951487	kg
arm_linear_mass	0.21605894	kg/m
frame_section_margin	0.0040176474	m
arm_section_margin	0.0055302728	m
structure_stress	1.1463086e+08	Pa
structure_stress_margin	5369144.6	Pa
structure_deflection	0.0038046413	m
structure_deflection_margin	0.0011953587	m
structure_buckling_margin	168.78473	N
empty_yaw_inertia	6.5554155	kg*m**2
empty_cg_x	0	m
empty_cg_z	0.36472713	m
loaded_yaw_inertia	19.542742	kg*m**2
loaded_cg_x	0	m
loaded_cg_z	0.48207878	m
half_yaw_inertia	13.049079	kg*m**2
half_cg_x	0.16778537	m
half_cg_z	0.44628068	m
loaded_sprung_mass	83.26408	kg
loaded_sprung_cg_x	0	m
loaded_sprung_cg_z	0.5346537	m
relocation_time	18.204569	s
relocation_energy	258.77176	J
relocation_peak_power	20.675213	W
fresh_site_margin	0.065489308	m
site_drive_margin	155.68393	N
site_turn_time	9.9545694	s
mobility_energy	2981.9286	J
drive_peak_power	93.251495	W
turn_peak_power	97.735448	W
ground_pressure_margin	11810.202	Pa
straight_motor_margin	4487.2477	N
turn_motor_margin	6656.9437	N
straight_traction_margin	210.28481	N
turn_traction_margin	10.204788	N
straight_power_margin	2611.888	W
turn_power_margin	2608.6595	W
mobility_stability_margin	0.27233423	m
trench_traverse_margin	0.1021989	m

Output port	Value	Native units
tip_stability_margin	0.048315408	m
tip_worst_mask	3	—
obstacle_pose_margin	0.17327174	m
obstacle_tip_margin	0.15338107	m
obstacle_pose_valid	1	—
obstacle_evaluated_height	0.4	m
obstacle_worst_wheel	0	—
obstacle_chassis_pitch	0.35711537	rad
obstacle_chassis_roll	0.34972659	rad
obstacle_loaded_cg_x	-0.11697912	m
obstacle_loaded_cg_y	-0.10742544	m
obstacle_loaded_cg_z	0.53858333	m
obstacle_drum_wheel_gap_bound	0.016584135	m
obstacle_drum_wheel_margin	0.0015841355	m
obstacle_clearance_pose_valid	1	—
obstacle_clearance_evaluated_height	0.4	m
slope_pitch_tip_margin	0.046671659	m
slope_roll_tip_margin	0.0067044852	m
slope_tip_margin	0.0067044852	m
slope_tip_angle	0.53396655	rad
rock_sweep_tip_margin	0.15338107	m
rock_sweep_worst_height	0.4	m
rock_sweep_raised_wheel	0	—
rock_sweep_pitch	0.35711537	rad
rock_sweep_roll	0.34972659	rad
rock_sweep_cg_x	-0.11697912	m
rock_sweep_cg_y	-0.10742544	m
rock_sweep_cg_z	0.53858333	m
rock_sweep_wheel_load_margin	1.374702	N
rock_sweep_load_height	0.4	m
rock_sweep_load_raised_wheel	2	—
rock_sweep_unloaded_wheel	2	—
rock_sweep_pose_valid	1	—
rock_sweep_evaluated_height	0.4	m
rock_sweep_samples	41	—
rock_sweep_force_balance_error	1.8474111e-13	N
plate_mass_total	0.74224856	kg
cf_tube_mass_total	0.085015267	kg
suspension_axle_mass	0.09482286	kg
suspension_bearing_span	0.34528514	m
cf_tube_outer_diameter	0.027400051	m

Output port	Value	Native units
plate_geometry_margin	0.037743485	m
frame_packing_margin	0.079388023	m
plate_geometry_valid	1	—
structure_cg_z	0.16343066	m
structure_sprung_cg_z	0.16581366	m
structure_yaw_inertia	0.037844371	kg*m**2
plate_stress	11142316	Pa
plate_stress_margin	1.0885768e+08	Pa
plate_buckling_margin	1027.9637	N
cf_bending_stress	1.0932793e+08	Pa
cf_bending_margin	65672065	Pa
frame_mount_relative_deflection	0.00049428859	m
rocker_deflection	0.0038046413	m
axle_deflection	0.00017946241	m
arm_deflection	3.5566042e-05	m
rocker_buckling_margin	5580.4547	N
arm_buckling_margin	168.78473	N
frame_gravity_actuator_force	1442.1201	N
frame_nominal_wheel_reaction	234.40821	N
frame_force_balance_error	5.1159077e-13	N
frame_load_solution_valid	1	—